For the use of a Registered Medical Practitioner or a Hospital or a Laboratory

GLIBETA TABLETS (5 mg + 500 mg)
(GLIBENCLAMIDE AND METFORMIN HYDROCHLORIDE TABLETS USP)

BRAND OR PRODUCT NAME

GLIBETA TABLETS (5 mg + 500 mg)

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

GLIBETA TABLETS (5 mg + 500 mg)
Each film coated tablet contains:
Glibenclamide USP…………………5 mg
Metformin Hydrochloride USP…….500 mg
Colours: Yellow oxide of Iron & Titanium Dioxide

PRODUCT DESCRIPTION

Yellow colored, capsule shaped, biconvex, film coated tablets debossed with “5” on one side and plain on other side.

DOSAGE FORM

Film coated tablet

CLINICAL PHARMACOLOGY

Pharmacodynamics

Glibeta tablet combines Glibenclamide and metformin hydrochloride, 2 antihyperglycemic agents with complementary mechanisms of action, to improve glycemic control in patients with type 2 diabetes.

Glibenclamide appears to lower blood glucose acutely by stimulating the release of insulin from the pancreas, an effect dependent upon functioning beta cells in the pancreatic islets. The mechanism by which Glibenclamide lowers blood glucose during long-term administration has not been clearly established. With chronic administration in patients with type 2 diabetes, the blood glucose-lowering effect persists despite a gradual decline in the insulin secretory response to the drug. Extra pancreatic effects may be involved in the mechanism of action of oral sulfonylurea hypoglycemic drugs.

Metformin hydrochloride is an antihyperglycemic agent that improves glucose tolerance in patients with type 2 diabetes, lowering both basal and postprandial plasma glucose. Metformin hydrochloride decreases hepatic glucose production, decreases intestinal absorption of glucose, and improves insulin sensitivity by increasing peripheral glucose uptake and utilization.

Pharmacokinetics

Absorption and Bioavailability

FDC Metformin and Glibenclamide

Following administration of a single FDC Metformin and Glibenclamide 5 mg/500 mg tablets with either a 20% glucose solution or a 20% glucose solution with food, there was no effect of food on the C_{max} and a relatively small effect of food on the AUC of the Glibenclamide component. The T_{max} for the Glibenclamide component was shortened from 7.5 hours to 2.75

hours with food compared to the same tablet strength administered fasting with a 20% glucose solution. The clinical significance of an earlier T_{max} for Glibenclamide after food is not known. The effect of food on the pharmacokinetics of the metformin component was indeterminate.

Glibenclamide

Mean serum levels of Glibenclamide, as reflected by areas under the serum concentration-time curve, increase in proportion to corresponding increases in dose. Bioequivalence has not been reported between FDC Metformin and Glibenclamide and single-ingredient Glibenclamide products.

Metformin Hydrochloride

The absolute bioavailability of a 500 mg metformin hydrochloride tablet given under fasting conditions is approximately 50% to 60%. Studies using single oral doses of metformin tablets of 500 mg and 1500 mg, and 850 mg to 2550 mg, indicate that there is a lack of dose proportionality with increasing doses, which is due to decreased absorption rather than an alteration in elimination. Food decreases the extent of and slightly delays the absorption of metformin, as shown by approximately a 40% lower peak concentration and a 25% lower AUC in plasma and a 35-minute prolongation of time to peak plasma concentration following administration of a single 850 mg tablet of metformin with food, compared to the same tablet strength administered fasting. The clinical relevance of these decreases is unknown.

<u>Distribution</u>

Glibenclamide

Sulfonylurea drugs are extensively bound to serum proteins. Displacement from protein binding sites by other drugs may lead to enhanced hypoglycemic action. *In vitro*, the protein binding exhibited by Glibenclamide is predominantly non-ionic, whereas that of other sulfonylureas (chlorpropamide, tolbutamide, tolazamide) is predominantly ionic. Acidic drugs, such as phenylbutazone, warfarin, and salicylates, displace the ionic-binding sulfonylureas from serum proteins to a far greater extent than the non-ionic binding Glibenclamide. It has not been shown that this difference in protein binding results in fewer drug-drug interactions with Glibenclamide tablets in clinical use.

Metformin Hydrochloride

The apparent volume of distribution (V/F) of metformin following single oral doses of 850 mg averaged 654±358 L. Metformin is negligibly bound to plasma proteins. Metformin partitions into erythrocytes, most likely as a function of time. At usual clinical doses and dosing schedules of metformin, steady state plasma concentrations of metformin are reached within 24 to 48 hours and are generally <1 μg/mL.

<u>Metabolism and Elimination</u>

Glibenclamide

The decrease of Glibenclamide in the serum of normal healthy individuals is biphasic; the terminal half-life is about 10 hours. The major metabolite of Glibenclamide is the 4 transhydroxy derivative. A second metabolite, the 3-cis-hydroxy derivative, also occurs. These metabolites probably contribute no significant hypoglycemic action in humans since they are only weakly active (1/400 and 1/40 as active, respectively, as Glibenclamide) in rabbits. Glibenclamide is excreted as metabolites in the bile and urine, approximately 50% by each route. This dual excretory pathway is qualitatively different from that of other sulfonylureas, which are excreted primarily in the urine.

Metformin Hydrochloride

Following oral administration, approximately 90% of the absorbed drug is eliminated via the renal route within the first 24 hours, with a plasma elimination half-life of approximately 6.2 hours. In blood, the elimination half-life is approximately 17.6 hours, suggesting that the erythrocyte mass may be a compartment of distribution.

Special Populations

Patients with Type 2 Diabetes

In the presence of normal renal function, there are no differences between single-or multipledose pharmacokinetics of metformin between patients with type 2 diabetes and normal subjects, nor is there any accumulation of metformin in either group at usual clinical doses.

Hepatic Insufficiency

No pharmacokinetic studies have been reported in patients with hepatic insufficiency for either Glibenclamide or metformin.

Renal Insufficiency

No information is available on the pharmacokinetics of Glibenclamide in patients with renal insufficiency.

In patients with decreased renal function (based on creatinine clearance), the plasma and blood half-life of metformin is prolonged and the renal clearance is decreased in proportion to the decrease in creatinine clearance.

Geriatrics

There is no information on the pharmacokinetics of Glibenclamide in elderly patients.

Pediatrics

There were no differences in pharmacokinetics of glibenclamide and metformin between pediatric patients and weight and gender matched healthy adults.

Gender

There is no information on the effect of gender on the pharmacokinetics of Glibenclamide.

Race

No information is available on race differences in the pharmacokinetics of Glibenclamide.
No studies of metformin pharmacokinetic parameters according to race have been reported.

INDICATION

FDC Metformin and Glibenclamide Tablets are indicated as an adjunct to diet and exercise to improve glycemic control in adults with type 2 diabetes mellitus.

RECOMMENDED DOSAGE AND MODE OF ADMINISTRATION

General Considerations

Dosage of FDC Metformin and Glibenclamide must be individualized on the basis of both effectiveness and tolerance while not exceeding the maximum recommended daily dose of 2000 mg metformin/20 mg glibenclamide. FDC Metformin and Glibenclamide should be given with meals with sufficiently high carbohydrate content to prevent the onset of hypoglycaemic episode. FDC Metformin and Glibenclamide should be initiated at a low dose, with gradual dose escalation as described below, in order to avoid hypoglycemia (largely due to glibenclamide), to reduce GI side effects (largely due to metformin), and to permit determination of the minimum effective dose for adequate control of blood glucose for the individual patient.

With initial treatment and during dose titration, appropriate blood glucose monitoring should be used to determine the therapeutic response to FDC Metformin and Glibenclamide and to identify the minimum effective dose for the patient. Thereafter, HbA1c (glycosylated haemoglobin) should be measured at intervals of approximately 3 months to assess the effectiveness of therapy. The therapeutic goal in all patients with type 2 diabetes is to decrease FPG, PPG, and HbA1c, to normal or as near normal as possible. Ideally, the response to therapy should be evaluated using HbA1c which is a better indicator of long-term glycaemic control than FPG alone.

No studies have been performed specifically examining the safety and efficacy of switching of FDC Metformin and Glibenclamide therapy in patients taking concomitant glibenclamide (or other sulphonylurea) plus metformin. Changes in glycaemic control may occur in such patients, with either hyperglycaemia or hypoglycaemia possible. Any change in therapy of type 2 diabetes should be undertaken with care and appropriate monitoring.

FDC Metformin and Glibenclamide Use in Previously Treated Patients (Second-Line Therapy)

Recommended starting dose: 500/2.5 mg or 500/5 mg twice daily with meals.

For patients not adequately controlled on either glibenclamide (or another sulphonylurea) or metformin alone, the recommended starting dose of FDC Metformin and Glibenclamide is 500/2.5 mg or 500/5 mg twice daily with the morning and evening meals. In order to avoid hypoglycaemia, the starting dose of FDC Metformin and Glibenclamide should not exceed the daily doses of glibenclamide or metformin already being taken. The daily dose should be titrated in increments of no more than 500/5 mg up to the minimum effective dose to achieve adequate control of blood glucose or to a maximum dose of 2000 mg/20 mg per day.

For patients previously treated with combination therapy of glibenclamide (or another sulphonylurea) plus metformin, if switched to FDC Metformin and Glibenclamide, the starting dose should not exceed the daily dose of glibenclamide (or equivalent dose of another sulphonylurea) and metformin already being taken. Patients should be monitored closely for signs and symptoms of hypoglycaemia following such a switch and the dose of FDC Metformin and Glibenclamide should be titrated as described above to achieve adequate control of blood glucose.

Specific Patient Populations

FDC Metformin and Glibenclamide are not recommended for use during pregnancy or for use in paediatric patients.

The initial and maintenance dosing of FDC Metformin and Glibenclamide should be conservative in patients with advanced age, due to the potential for decreased renal function in this population. Any dosage adjustment requires a careful assessment of renal function. Generally, elderly, debilitated, and malnourished patients should not be titrated to the maximum dose of FDC Metformin and Glibenclamide to avoid the risk of hypoglycaemia.

Monitoring of renal function is necessary to aid in prevention of metformin-associated lactic acidosis, particularly in the elderly.

Renal Impairment:

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

The maximum daily dose of metformin should preferably be divided into 2-3 daily doses. Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin in patients with GFR <60 ml/min.

If no adequate strength of Glibeta is available, individual monocomponents should be used instead of the fixed dose combination.

GFR mL/min	**Metformin**	**[other monocomponent]**
60-89	Maximum daily dose is 3000 mg. Dose reduction may be considered in relation to declining renal function.	
45-59	Maximum daily dose is 2000 mg. The starting dose is at most half of the maximum dose.	
30-44	Maximum daily dose is 1000 mg. The starting dose is at most half of the maximum dose.	
<30	Metformin is contraindicated.	

CONTRAINDICATIONS:

FDC Metformin and Glibenclamide is contraindicated in patients with:

1. Renal disease or renal dysfunction (e.g., as suggested by serum creatinine levels ≥1.5 mg/dL [males], ≥1.4 mg/dL [females], or abnormal creatinine clearance) which may also result from conditions such as cardiovascular collapse (shock), acute myocardial infarction, and septicemia.

2. Known hypersensitivity to metformin hydrochloride or Glibenclamide.

3. Acute or chronic metabolic acidosis, including diabetic ketoacidosis, with or without coma. Diabetic ketoacidosis should be treated with insulin.

4. Concomitant administration of bosentan.

5. Severely reduced kidney function (GFR <30 mL/min)

6. Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

FDC Metformin and Glibenclamide should be temporarily discontinued in patients undergoing radiologic studies involving intravascular administration of iodinated contrast materials, because use of such products may result in acute alteration of renal function.

SPECIAL WARNNG AND PRECAUTION FOR USE
Metformin Hydrochloride

Lactic acidosis:
Lactic acidosis is a rare, but serious, metabolic complication that can occur due to metformin accumulation during treatment with FDC Metformin and Glibenclamide (Glibenclamide and Metformin HCl) Tablets; when it occurs, it is fatal in approximately 50% of cases. Lactic acidosis may also occur in association with a number of pathophysiologic conditions, including diabetes mellitus, and whenever there is significant tissue hypoperfusion and hypoxemia. Lactic acidosis is characterized by elevated blood lactate levels (>5 mmol/L), decreased blood pH, electrolyte disturbances with an increased anion gap, and an increased lactate/pyruvate ratio. When metformin is implicated as the cause of lactic acidosis, metformin plasma levels >5 μg/mL are generally found.

Patients with congestive heart failure requiring pharmacologic management, in particular those with unstable or acute congestive heart failure who are at risk of hypoperfusion and hypoxemia, are at increased risk of lactic acidosis. The risk of lactic acidosis increases with the degree of renal dysfunction and the patient's age. The risk of lactic acidosis may, therefore, be significantly decreased by regular monitoring of renal function in patients taking metformin and by use of the minimum effective dose of metformin. In particular, treatment of the elderly should be accompanied by careful monitoring of renal function. FDC Metformin and Glibenclamide treatment should not be initiated in patient's ≥80 years of age unless measurement of creatinine clearance demonstrates that renal function is not reduced, as these patients are more susceptible to developing lactic acidosis. In addition, FDC Metformin and Glibenclamide should be promptly withheld in the presence of any condition associated with hypoxemia, dehydration, or sepsis. Because impaired hepatic function may significantly limit the ability to clear lactate, FDC Metformin and Glibenclamide should generally be avoided in patients with clinical or laboratory evidence of hepatic disease. Patients should be cautioned against excessive alcohol intake, either acute or chronic, when taking FDC Metformin and Glibenclamide, since alcohol potentiates the effects of metformin hydrochloride on lactate metabolism. In addition, FDC Metformin and Glibenclamide should be temporarily discontinued prior to any intravascular radiocontrast study and for any surgical procedure. The onset of lactic acidosis often is subtle, and accompanied only by nonspecific symptoms such as malaise, myalgias, respiratory distress, increasing somnolence, and nonspecific abdominal distress. There may be associated hypothermia,hypotension and resistant bradyarrhythmias with more marked acidosis. The patient and the patient's physician must be aware of the possible importance of such symptoms and the patient should be instructed to notify the physician immediately if they occur. FDC Metformin and Glibenclamide should be withdrawn until the situation is clarified. Serum electrolytes, ketones, blood glucose, and if indicated, blood pH, lactate levels, and even blood metformin levels may be useful. Once a patient is stabilized on any dose level of FDC Metformin and Glibenclamide, gastrointestinal symptoms, which are common during initiation of therapy with metformin, are unlikely to be drug related. Later occurrence of gastrointestinal symptoms could be due to lactic acidosis or other serious disease.

Levels of fasting venous plasma lactate above the upper limit of normal but less than 5 mmol/L in patients taking FDC Metformin and Glibenclamide do not necessarily indicate impending lactic acidosis and may be explainable by other mechanisms, such as poorly controlled diabetes or obesity, vigorous physical activity, or technical problems in sample handling.

Lactic acidosis should be suspected in any diabetic patient with metabolic acidosis lacking evidence of ketoacidosis (ketonuria and ketonemia).

Lactic acidosis is a medical emergency that must be treated in a hospital setting. In a patient with lactic acidosis who is taking FDC Metformin and Glibenclamide, the drug should be discontinued immediately and general supportive measures promptly instituted. Because metformin hydrochloride is dialyzable (with a clearance of up to 170 mL/min under good hemodynamic conditions), prompt hemodialysis is recommended to correct the acidosis and remove the accumulated metformin. Such management often results in prompt reversal of symptoms and recovery.

Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.

In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.

Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH(< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Renal function:

GFR should be assessed before treatment initiation and regularly there after [See Section Recommended Dosage]. Metformin is contraindicated in patients with GFR <30 mL/min and should be temporarily discontinued in the presence of conditions that alter renal function [See Section Contraindications].

Special warning on increased risk of cardiovascular mortality
The administration of oral hypoglycemic drugs has been reported to be associated with increased cardiovascular mortality as compared to treatment with diet alone or diet plus insulin.

PRECAUTIONS

General

Macrovascular Outcomes

There have been no clinical studies reporting conclusive evidence of macrovascular risk reduction with FDC Metformin and Glibenclamide or any other antidiabetic drug.

FDC Metformin and Glibenclamide

Hypoglycemia

FDC Metformin and Glibenclamide is capable of producing hypoglycemia or hypoglycemic symptoms, therefore, proper patient selection, dosing, and instructions are important to avoid potential hypoglycemic episodes. The risk of hypoglycemia is increased when caloric intake is deficient, when strenuous exercise is not compensated by caloric supplementation or during concomitant use with other glucose-lowering agents or ethanol. Renal or hepatic insufficiency may cause elevated drug levels of both Glibenclamide and metformin hydrochloride, and the hepatic insufficiency may also diminish gluconeogenic capacity, both of which increase the risk of hypoglycemic reactions. Elderly, debilitated, or malnourished patients and those with adrenal or pituitary insufficiency or alcohol intoxication are particularly susceptible to hypoglycemic effects. Hypoglycemia may be difficult to recognize in the elderly and people who are taking beta-adrenergic blocking drugs.

Glibenclamide

Hemolytic anemia

Treatment of patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency with sulfonylurea agents can lead to hemolytic anemia. Because FDC Metformin and Glibenclamide belongs to the class of sulfonylurea agents, caution should be used in patients with G6PD deficiency and a non-sulfonylurea alternative should be considered. In postmarketing reports, hemolytic anemia has also been reported in patients who did not have known G6PD deficiency.

Metformin Hydrochloride

Monitoring of renal function

Metformin is known to be substantially excreted by the kidney and the risk of metformin accumulation and lactic acidosis increases with the degree of impairment of renal function. Thus, patients with serum creatinine levels above the upper limit of normal for their age should not receive FDC Metformin and Glibenclamide. In patients with advanced age, FDC Metformin and Glibenclamide should be carefully titrated to establish the minimum dose for adequate glycemic effect, because aging is associated with reduced renal function. In elderly patients, particularly those ≥ 80 years of age, renal function should be monitored regularly and, generally, FDC Metformin and Glibenclamide should not be titrated to the maximum dose. Before initiation of FDC Metformin and Glibenclamide therapy and at least annually thereafter, renal function should be assessed and verified as normal. In patients in whom development of renal dysfunction is anticipated, renal function should be assessed more frequently and FDC Metformin and Glibenclamide discontinued if evidence of renal impairment is present.

Use of concomitant medications that may affect renal function or metformin disposition

Concomitant medication(s) that may affect renal function or result in significant hemodynamic change or may interfere with the disposition of metformin, such as cationic drugs that are eliminated by renal tubular secretion, should be used with caution.

Radiologic studies involving the use of intravascular iodinated contrast materials (for example, intravenous urogram, intravenous cholangiography, angiography, and computed

tomography (CT) scans with intravascular contrast materials)
Intravascular contrast studies with iodinated materials can lead to acute alteration of renal function and have been associated with lactic acidosis in patients receiving metformin. Therefore, in patients in whom any such study is planned, FDC Metformin and Glibenclamide should be temporarily discontinued at the time of or prior to the procedure, and withheld for 48 hours subsequent to the procedure and reinstituted only after renal function has been reevaluated and found to be normal.

Hypoxic states
Cardiovascular collapse (shock) from whatever cause, acute congestive heart failure, acute myocardial infarction, and other conditions characterized by hypoxemia have been associated with lactic acidosis and may also cause prerenal azotemia. When such events occur in patients on FDC Metformin and Glibenclamide therapy, the drug should be promptly discontinued.

Surgical procedures
FDC Metformin and Glibenclamide therapy should be temporarily suspended for any surgical procedure (except minor procedures not associated with restricted intake of food and fluids) and should not be restarted until the patient’s oral intake has resumed and renal function has been evaluated as normal.

Alcohol intake
Alcohol is known to potentiate the effect of metformin on lactate metabolism. Patients, therefore, should be warned against excessive alcohol intake, acute or chronic, while receiving FDC Metformin and Glibenclamide. Due to its effect on the gluconeogenic capacity of the liver, alcohol may also increase the risk of hypoglycemia.

Impaired hepatic function
Since impaired hepatic function has been associated with some cases of lactic acidosis, FDC Metformin and Glibenclamide should generally be avoided in patients with clinical or laboratory evidence of hepatic disease.

Vitamin B_{12} levels
A decrease to subnormal levels of previously normal serum vitamin B_{12}, without clinical manifestations, was reported. Such decrease, possibly due to interference with B_{12} absorption from the B_{12}-intrinsic factor complex is, however, very rarely associated with anemia and appears to be rapidly reversible with discontinuation of metformin or vitamin B_{12} supplementation. Measurement of hematologic parameters on an annual basis is advised in patients on metformin and any apparent abnormalities should be appropriately investigated and managed.

Certain individuals (those with inadequate vitamin B_{12} or calcium intake or absorption) appear to be predisposed to developing subnormal vitamin B_{12} levels. In these patients, routine serum vitamin B_{12} measurements at 2- to 3-year intervals may be useful.

Change in clinical status of patients with previously controlled type 2 diabetes
A patient with type 2 diabetes previously well controlled on metformin who develops laboratory abnormalities or clinical illness (especially vague and poorly defined illness) should be evaluated promptly for evidence of ketoacidosis or lactic acidosis. Evaluation should include serum electrolytes and ketones, blood glucose and, if indicated, blood pH, lactate, pyruvate,

and metformin levels. If acidosis of either form occurs, FDC Metformin and Glibenclamide must be stopped immediately and other appropriate corrective measures initiated.

Addition of Thiazolidinediones to FDC Metformin and Glibenclamide Therapy
Hypoglycemia
Patients receiving FDC Metformin and Glibenclamide in combination with a thiazolidinedione may be at risk for hypoglycemia.

Weight gain
Weight gain was seen with the addition of rosiglitazone to FDC Metformin and Glibenclamide, similar to that reported for thiazolidinedione therapy alone.

Hepatic effects
When a thiazolidinedione is used in combination with FDC Metformin and Glibenclamide, periodic monitoring of liver function tests should be performed in compliance with the labeled recommendations for the thiazolidinedione.

Laboratory Tests
Periodic fasting blood glucose (FBG) and HbA1c measurements should be performed to monitor therapeutic response.
Initial and periodic monitoring of hematologic parameters (eg, hemoglobin/hematocrit and red blood cell indices) and renal function (serum creatinine) should be performed, at least on an annual basis. While megaloblastic anemia has rarely been seen with metformin therapy, if this is suspected, vitamin B_{12} deficiency should be excluded.

INTERACTIONS WITH OTHER MEDICAMENTS AND OTHER FORMS OF INTERACTION
FDC Metformin and Glibenclamide
Certain drugs tend to produce hyperglycemia and may lead to loss of blood glucose control. These drugs include thiazides and other diuretics, corticosteroids, phenothiazines, thyroid products, estrogens, oral contraceptives, phenytoin, nicotinic acid, sympathomimetics, calcium channel blocking drugs, and isoniazid. When such drugs are administered to a patient receiving FDC Metformin and Glibenclamide, the patient should be closely observed for loss of blood glucose control. When such drugs are withdrawn from a patient receiving FDC Metformin and Glibenclamide, the patient should be observed closely for hypoglycemia. Metformin is negligibly bound to plasma proteins and is therefore, less likely to interact with highly protein-bound drugs such as salicylates, sulfonamides, chloramphenicol, and probenecid as compared to sulfonylureas, which are extensively bound to serum proteins.

Glibenclamide
The hypoglycemic action of sulfonylureas may be potentiated by certain drugs, including nonsteroidal anti-inflammatory agents and other drugs that are highly protein bound, salicylates, sulfonamides, chloramphenicol, probenecid, coumarins, monoamine oxidase inhibitors, and beta-adrenergic blocking agents. When such drugs are administered to a patient receiving FDC Metformin and Glibenclamide, the patient should be observed closely for hypoglycemia. When such drugs are withdrawn from a patient receiving FDC Metformin and Glibenclamide, the patient should be observed closely for loss of blood glucose control. An increased risk of liver enzyme elevations was observed in patients receiving Glibenclamide concomitantly with bosentan. Therefore concomitant administration of FDC Metformin and Glibenclamide and bosentan is contraindicated.

A possible interaction between Glibenclamide and ciprofloxacin, a fluoroquinolone antibiotic, has been reported, resulting in a potentiation of the hypoglycemic action of Glibenclamide. The mechanism for this interaction is not known.
A potential interaction between oral miconazole and oral hypoglycemic agents leading to severe hypoglycemia has been reported. Whether this interaction also occurs with the intravenous, topical, or vaginal preparations of miconazole is not known.

Colesevelam: Concomitant administration of colesevelam and Glibenclamide resulted in reductions in Glibenclamide AUC and C_{max}.

Metformin Hydrochloride
Furosemide
Furosemide increased the metformin plasma and blood C_{max} and AUC, without any significant change in metformin renal clearance. No information is available about the interaction of metformin and furosemide when co-administered chronically.

Nifedipine
Co-administration of nifedipine increased plasma metformin C_{max} and AUC and increased the amount excreted in the urine. T_{max} and half-life were unaffected. Nifedipine appears to enhance the absorption of metformin. Metformin had minimal effects on nifedipine.

Cationic drugs
Cationic drugs (e.g., amiloride, digoxin, morphine, procainamide, quinidine, quinine, ranitidine, triamterene, trimethoprim, or vancomycin) that are eliminated by renal tubular secretion theoretically have the potential for interaction with metformin by competing for common renal tubular transport systems. Metformin had no effect on cimetidine pharmacokinetics. Although such interactions remain theoretical (except for cimetidine), careful patient monitoring and dose adjustment of FDC Metformin and Glibenclamide and/or the interfering drug is recommended in patients who are taking cationic medications that are excreted via the proximal renal tubular secretory system.

Other
The pharmacokinetics of metformin and propranolol and metformin and ibuprofen were not affected when co-administered.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION
Pregnancy
Adequate blood glucose control allows pregnancy to proceed normally in this category of patients. Fixed dose combination of Glibenclamide and Metformin must not be used for the treatment of diabetes during pregnancy. It is imperative that insulin be used to achieve adequate blood glucose control. It is recommended that the patient be transferred from oral antidiabetic therapy to insulin as soon as she plans to become pregnant or if pregnancy is exposed to this medicinal product. Neonatal blood glucose monitoring is recommended. In humans, in the absence of data concerning passage of metformin and glibenclamide into breast milk, and in view of the risk of neonatal hypoglycaemia, this medicinal product is contraindicated in the event of breast-feeding.

Nursing Mothers

Although it is not known whether Glibenclamide is excreted in human milk, some sulfonylurea drugs are known to be excreted in human milk. Studies in lactating rats reported that metformin is excreted into milk and reaches levels comparable to those in plasma. Similar studies have not been reported in nursing mothers. Because the potential for hypoglycemia in nursing infants may exist, a decision should be made whether to discontinue nursing or to discontinue FDC Metformin and Glibenclamide, taking into account the importance of the drug to the mother. If FDC Metformin and Glibenclamide is discontinued, and if diet alone is inadequate for controlling blood glucose, insulin therapy should be considered.

Geriatric Use

Metformin hydrochloride is known to be substantially excreted by the kidney and because the risk of serious adverse reactions to the drug is greater in patients with impaired renal function, FDC Metformin and Glibenclamide should only be used in patients with normal renal function. Because aging is associated with reduced renal function, FDC Metformin and Glibenclamide should be used with caution as age increases. Care should be taken in dose selection and should be based on careful and regular monitoring of renal function. Generally, elderly patients should not be titrated to the maximum dose of FDC Metformin and Glibenclamide.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS

FDC METFORMIN AND GLIBENCLAMIDE

The percent of patients reporting events and types of adverse events of FDC Metformin and Glibenclamide (all strengths) as initial therapy and second-line therapy are listed below.

Most Common Adverse Events of FDC Metformin and Glibenclamide Used as Initial or Second-Line Therapy

Adverse Event

Upper respiratory infection, diarrhea, headache, Nausea/vomiting, abdominal pain, dizziness. Disulfiram-like reactions have very rarely been reported in patients treated with Glibenclamide tablets.

Hypoglycemia

The incidence of reported symptoms of hypoglycemia. (such as dizziness, shakiness, sweating, and hunger)

Gastrointestinal Reactions

Diarrhea, nausea/vomiting, and abdominal pain were the most common adverse events with FDC Metformin and Glibenclamide and were more frequent at higher dose levels.

In post marketing reports cholestatic jaundice and hepatitis may occur rarely which may progress to liver failure; FDC Metformin and Glibenclamide should be discontinued if this occurs.

OVERDOSE AND TREATMENT

Glibenclamide

Overdosage of sulfonylureas, including Glibenclamide tablets, can produce hypoglycemia.
Mild hypoglycemic symptoms, without loss of consciousness or neurological findings, should be treated aggressively with oral glucose and adjustments in drug dosage and/or meal patterns. Close monitoring should continue until the physician is assured that the patient is out of danger. Severe hypoglycemic reactions with coma, seizure, or other neurological impairment occur infrequently, but constitute medical emergencies requiring immediate hospitalization. If

hypoglycemic coma is diagnosed or suspected, the patient should be given a rapid intravenous injection of concentrated (50%) glucose solution. This should be followed by a continuous infusion of a more dilute (10%) glucose solution at a rate that will maintain the blood glucose at a level above 100 mg/dL. Patients should be closely monitored for a minimum of 24 to 48 hours, since hypoglycemia may recur after apparent clinical recovery.

Metformin Hydrochloride

Overdose of metformin hydrochloride has occurred, including ingestion of amounts >50 g. Hypoglycemia was reported in approximately 10% of cases, but no causal association with metformin hydrochloride has been established. Lactic acidosis has been reported in approximately 32% of metformin overdose cases. Metformin is dialyzable with a clearance of up to 170 mL/min under good hemodynamic conditions. Therefore, hemodialysis may be useful for removal of accumulated drug from patients in whom metformin overdosage is suspected.

STORAGE CONDITIONS

Store below 30°C.

PACKAGING AVAILABLE

Glibeta Tablets (5 mg + 500 mg) are packed in Alu- Alu blister of 10 tablets. Such blisters containing 10 tablets are packed into boxes of 10’s, 30’s, 100’s and 120’s. Not all presentations or all pack sizes may be marketed.

DATE OF REVISION OF PACKAGE INSERT

November 1st, 2017

NAME AND ADDRESS OF MANUFACTURER

torrent PHARMA

TORRENT PHARMACEUTICALS LTD.
Indrad 382 721, District: Mehsana
INDIA.